

Frédéric Andriantsehero

AI Engineer · Agentic AI · LLMs · RAG

Bordeaux, France

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Summary

AI Engineer specializing in Agentic AI, Large Language Models and Retrieval-Augmented Generation, currently developing an end-to-end AI agent architecture at Inria Bordeaux. Hands-on experience building AI-powered applications across the stack, from RAG pipelines and Python/FastAPI backend services to interactive web interfaces, using modern software engineering tools including Docker and Git. Currently completing a French Engineering Degree (Master's level) in AI and Cognitive Science, graduating in 2026.

Experience

Inria (National Institute for Research in Digital Science and Technology)

Bordeaux, France

AI Research Engineer Intern – Agentic AI

Mar. 2026 - Present

- Engineered an AI agent from first principles without relying on agentic frameworks, supporting and accelerating PhD research workflows.
- Designed and implemented an end-to-end RAG pipeline covering document ingestion, chunking, embedding, vector indexing and retrieval, integrated with agent tool-use for multi-step task execution.
- Built an interactive web interface enabling researchers to interact with the agent and its tools.
- Benchmarked and evaluated locally deployed open-source LLMs across performance, latency and task quality.

Karunya University

Coimbatore, Tamil Nadu,
India

Reinforcement Learning Research Intern

Jun. 2025 - Sep. 2025

- Designed and implemented two distinct navigation approaches for an autonomous indoor hospital robot: a classical deterministic path-planning algorithm and a reinforcement learning-based navigation model.
- Conducted a comparative analysis between the algorithmic and reinforcement learning models.
- Successfully deployed and optimized the reinforcement learning model onto an embedded Raspberry Pi.

Skills

AI/GenAI	Agentic AI, LLMs, RAG, OpenAI API, LangChain, ChromaDB
Software Engineering	Next.js, React, Node.js, MongoDB, FastAPI, REST APIs, SQL, Docker, Git
Programming	Python, Java, C#
Machine Learning	Deep Learning, Reinforcement Learning, NLP, PyTorch, NumPy, Pandas
Languages	French (native), English (IELTS 7.5), Spanish (basic)

Education

ENSC (École Nationale Supérieure de Cognitique (National Graduate School of Cognitive Engineering))

Bordeaux, France

Engineer in AI and Cognitive Science (Master's level)

2023 - 2026

- Acquired a strong foundational background in general engineering, computer science, and signal processing, with a core focus on cognitive sciences.
- Specialized in Artificial Intelligence during the final year, focusing on advanced topics such as NLP, Large Language Models (LLMs), Reinforcement Learning (RL), and Deep Learning architectures (RNNs).

Classes Préparatoires aux Grandes Écoles (CPGE)

Brest, France

Two-year Intensive Undergraduate Program in Mathematics and Physics (equivalent of a Bachelor degree)

2021 - 2023

- Acquired a rigorous and high-level foundation in advanced Mathematics, Physics, Computer Science, and Engineering Sciences.